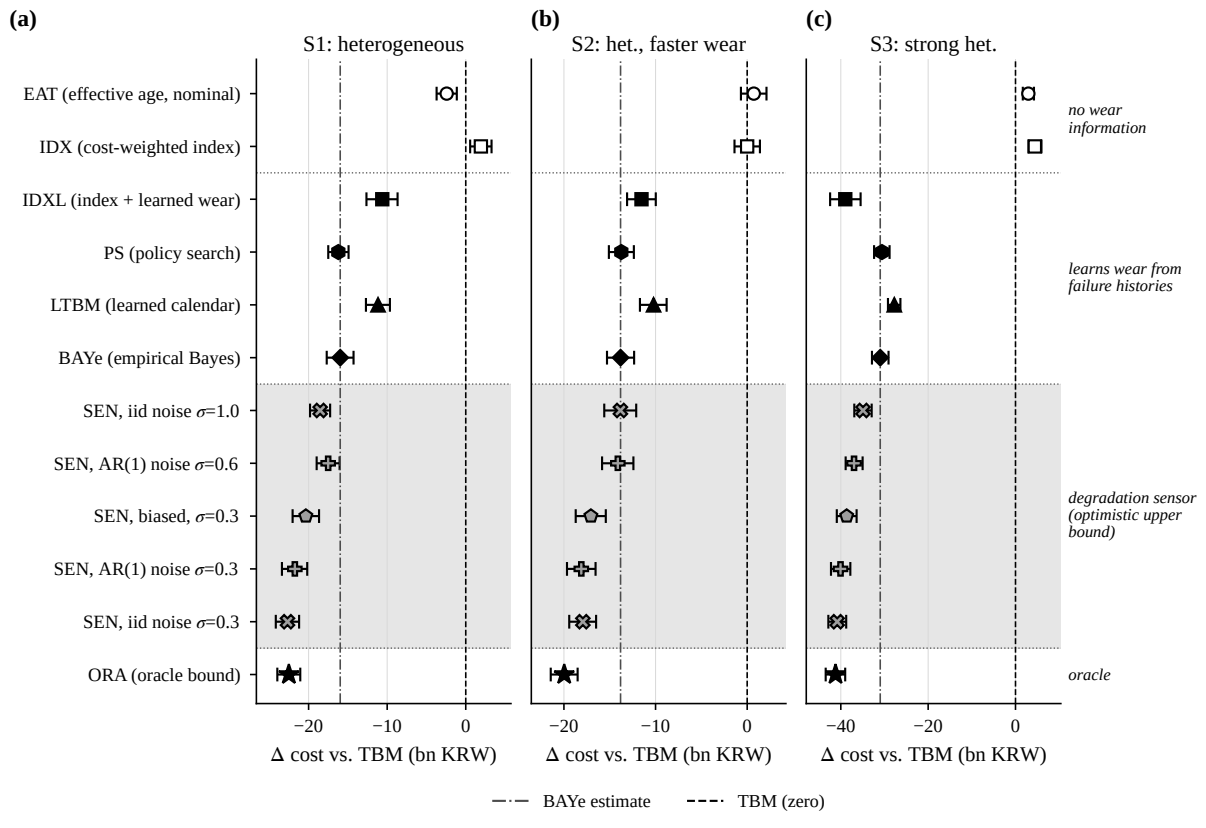
(a)
(b)
(c)
S1: heterogeneous
S2: het., faster wear
S3: strong het.
EAT (effective age, nominal)
IDX (cost-weighted index)
IDXL (index + learned wear)
PS (policy search)
LTBM (learned calendar)
BAYe (empirical Bayes)
SEN, iid noise σ=1.0
SEN, AR(1) noise σ=0.6
SEN, biased, σ=0.3
SEN, AR(1) noise σ=0.3
SEN, iid noise σ=0.3
ORA (oracle bound)
no wear information
learns wear from failure histories
degradation sensor (optimistic upper bound)
oracle
−20
−10
0
−40
−20
0
Δ cost vs. TBM (bn KRW)
Δ cost vs. TBM (bn KRW)
Δ cost vs. TBM (bn KRW)
BAYe estimate
TBM (zero)